fused ring and bridged ring;

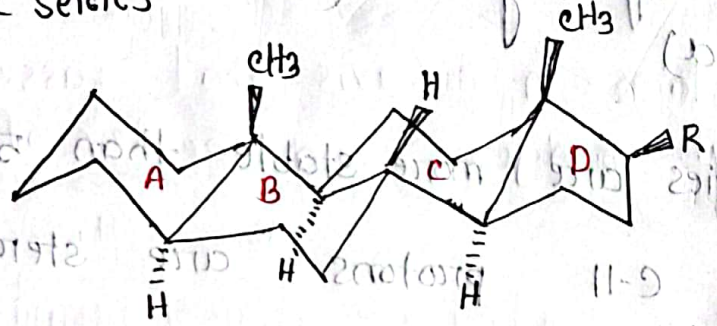
final-2023

Alamin-450

a. Draw the conformational structure of [5α series] and [5β series] of steroid molecule.

Ans:

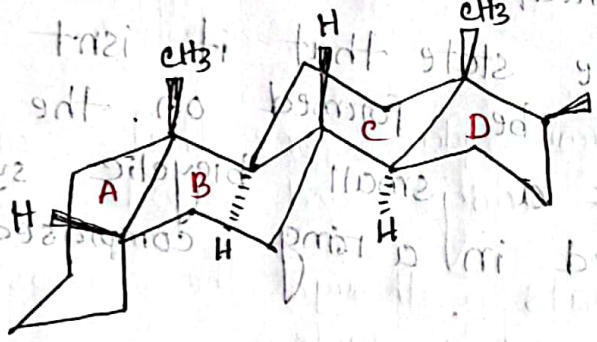
5α series



Ring A,B are fused in trans form, B-C trans form,

C,D trans

5β series



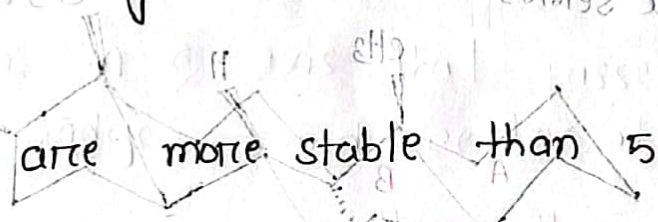
Ring A,B are in cis form and B,C-trans. also C,D trans in manner.

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(b) mention the features that are common in naturally occuring steroid molecules.

Ans:

1. All ring fusions are trans because of stability
2. many steroids have methyl groups called angular methyl at 10 and 13 position
3. Groups or atoms which are same as 10-CH_3 & 13-CH_3 like up they are (β) and pointing downward (α)
4. 5α series are more stable than 5β series.
5. C-1 and C-11 protons are sterically hindered



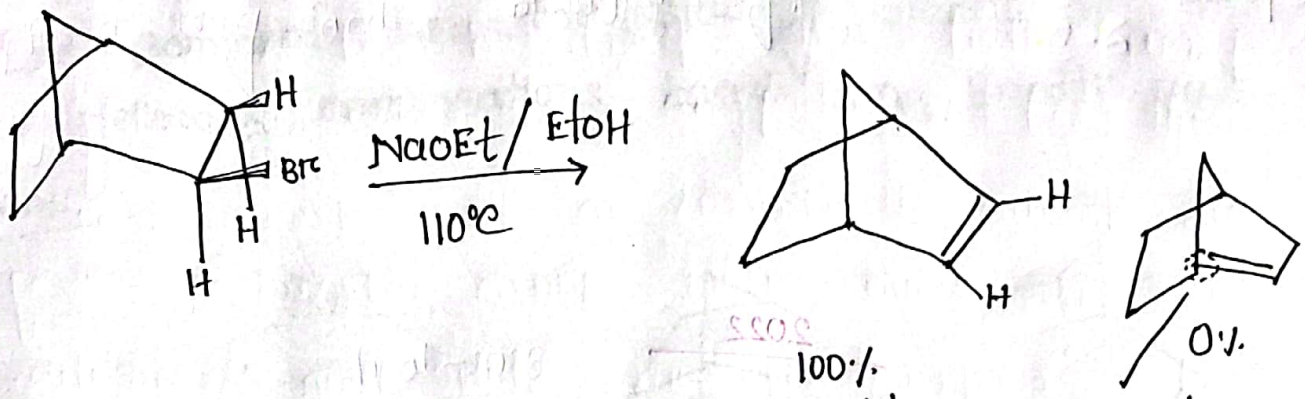
(c)

Q. State Bredt's rule with example.

Ans: The rule state that it isn't possible for double bonds to be formed on the bridgehead carbon atoms of a small bicyclic system if it is being incorporated in a ring comprised of fewer than 8 carbon.

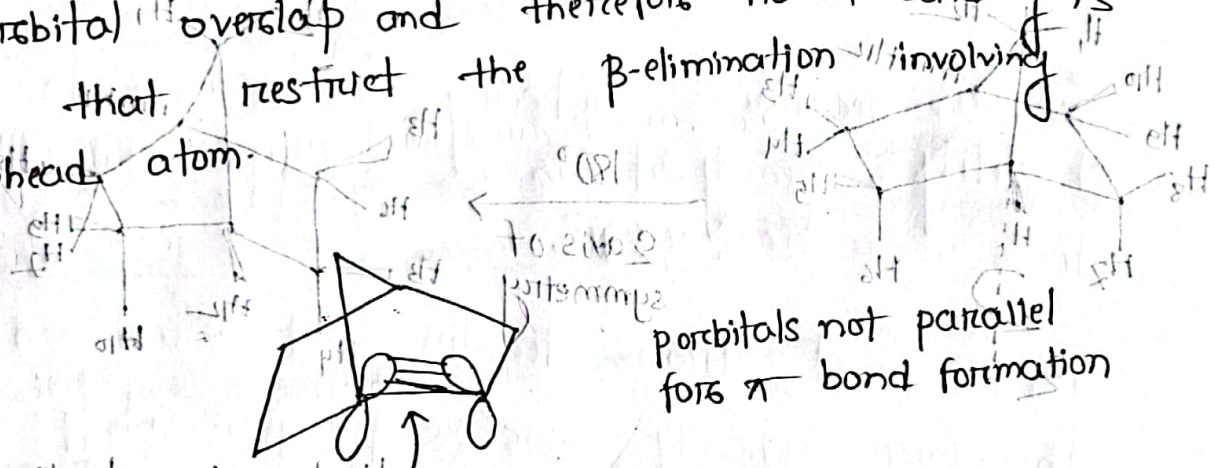
For example during β-elimination in bicyclic system the bridgehead atoms are not involved due to unfavourable angular and torsional strain.

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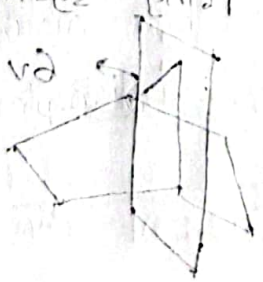
Q. Explain that during β- elimination rxn in bycic compounds the bridgehead atoms arent involved

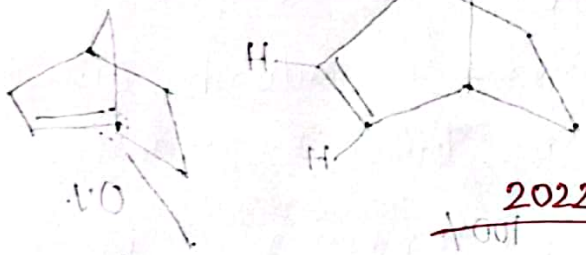
ANS: It is due to the sp^2 carbon at bridgehead position have a p-orbital that is oriented roughly at 90° relative to the p orbital at an adjacent sp^2 carbon, resulting poors orbital overlap and therefore no π bonding is possible. that restrict the β-elimination involving bridgehead atom.



p orbitals not parallel for π bond formation

insufficient overlap





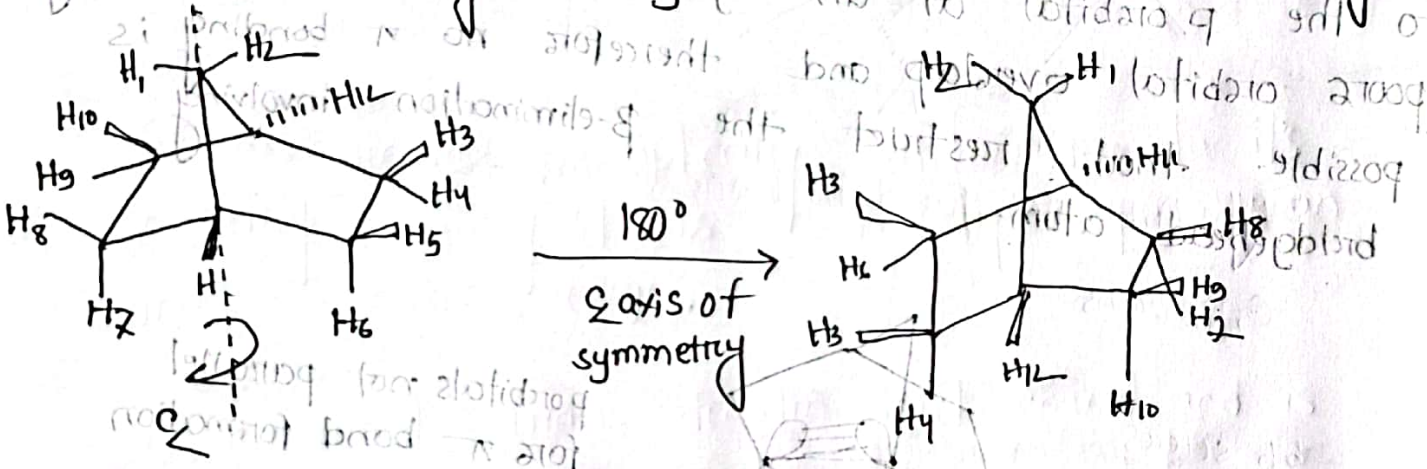
2022

Explain that Norborane has point group C_{2v}

Ans:

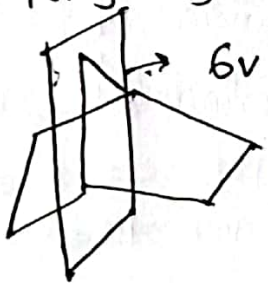
Norborane is a bicyclic compound consist of a six-membered ring fused with a one-carbon ring across the ring.

Norborane contains a C_2 axis that passes the molecule through bridge and center of the ring



Also two mirror planes one that bisect the molecule through the bridge and center of the ring

Another that cuts through the opposite plane due to its symmetrical ring system.

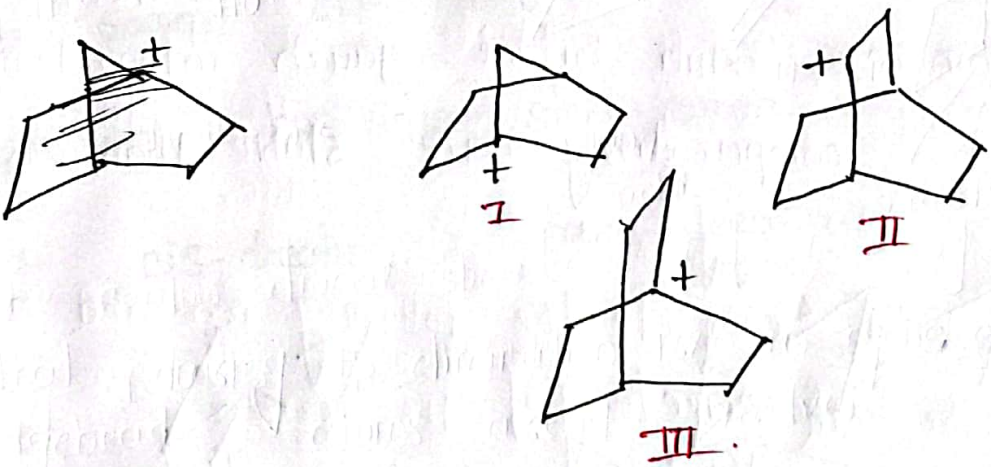


Hence norborane has point group C_{2v}

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* Comment on the stability orders of the following carbocations based on Bredt's rule—

Ans:



• Bredt's rule states that bridgehead carbocation (or double bonds) are unstable in small ring system because the geometry doesn't allow proper P-orbital overlap.

• Here the first and third structure have bridgehead carbocation violating Bredt's rule

The 2nd structure is accompanied with Bredt's rule.

• In the case of compound III there has 8 membered cyclic compound has comparatively less unstability.

So the Stability orders would be

II > III > I

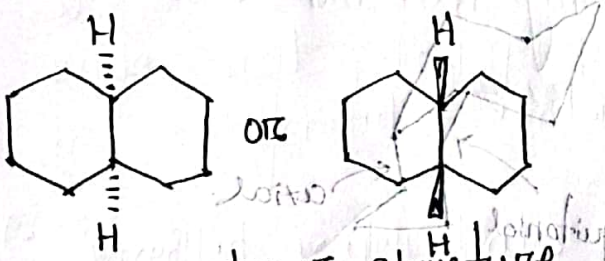
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F-2021, 2018

Q. Draw the structure of different stereoisomers of decalin. comment on their chirality and stability.

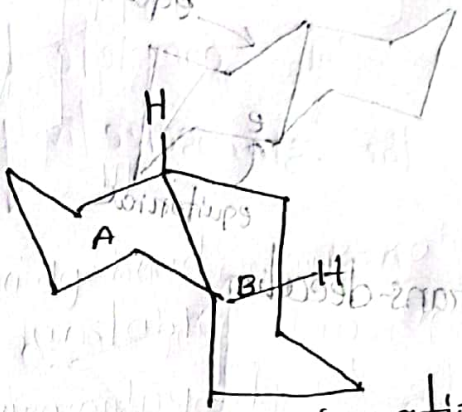
Ans: Decalin has two stereo isomers :—

1 cis-decalin 2. trans-decalin.

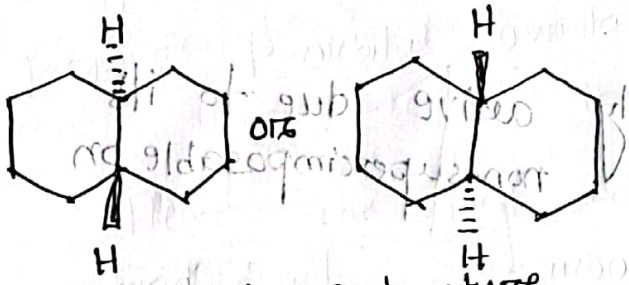


planar structure

cis decalin

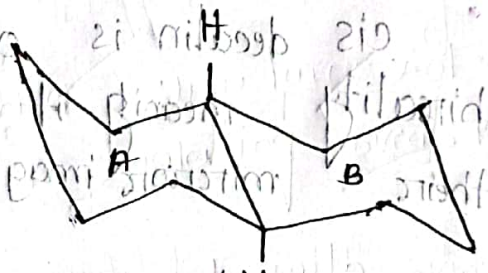


chair conformation



planar structure

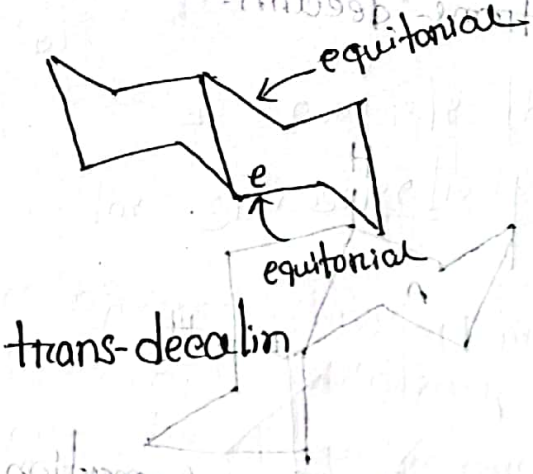
trans-decalin



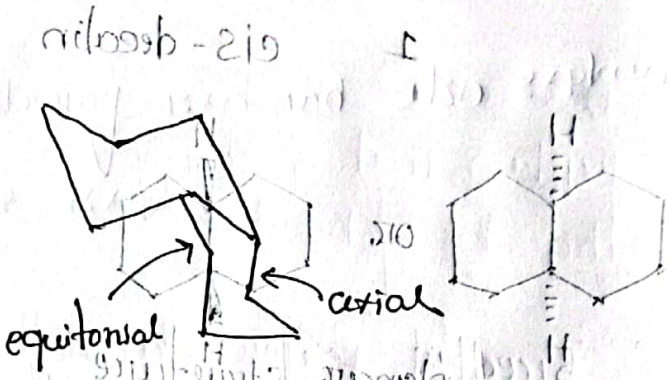
chair conformation.

- trans decalin is more stable than cis decalin since all carbons are equitorial, there is no 1,3-diaxial interaction on the others hand in cis-decalin, one carbon is equitorial and the others is axial that leads to four 1,3-diaxial interaction making molecule unstable. Also in cis decalin there is extra 3-butane gauche interaction that

means steric interaction betn non-bonded atoms or groups that are positioned at a dihedral angle (60°) resulting in extra unstability in cis decalin, whereas in trans-decalin that minimize extra butane gauch interaction making it comperatively more stable than cis-decalin.



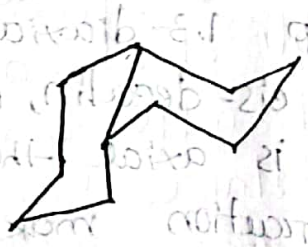
trans-decalin



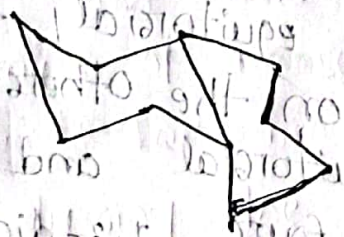
cis-decalin.

cis decalin is optically active due to its chirality meanig it is non-superimposable on their mirror image.

In contrast, trans decalin isn't chiral and is therefore optically inactive because it possesses a plane of symmetry.



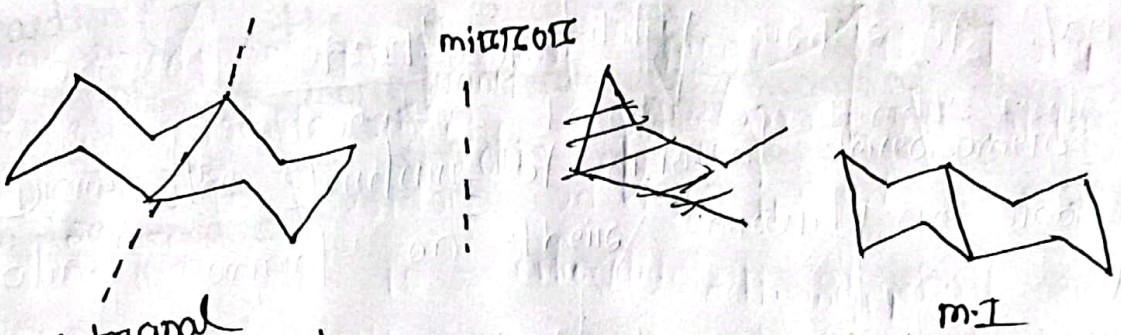
cis decalin



mirror image

form a pairs of non-resolveable enantiomers

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internal plane of symmetry

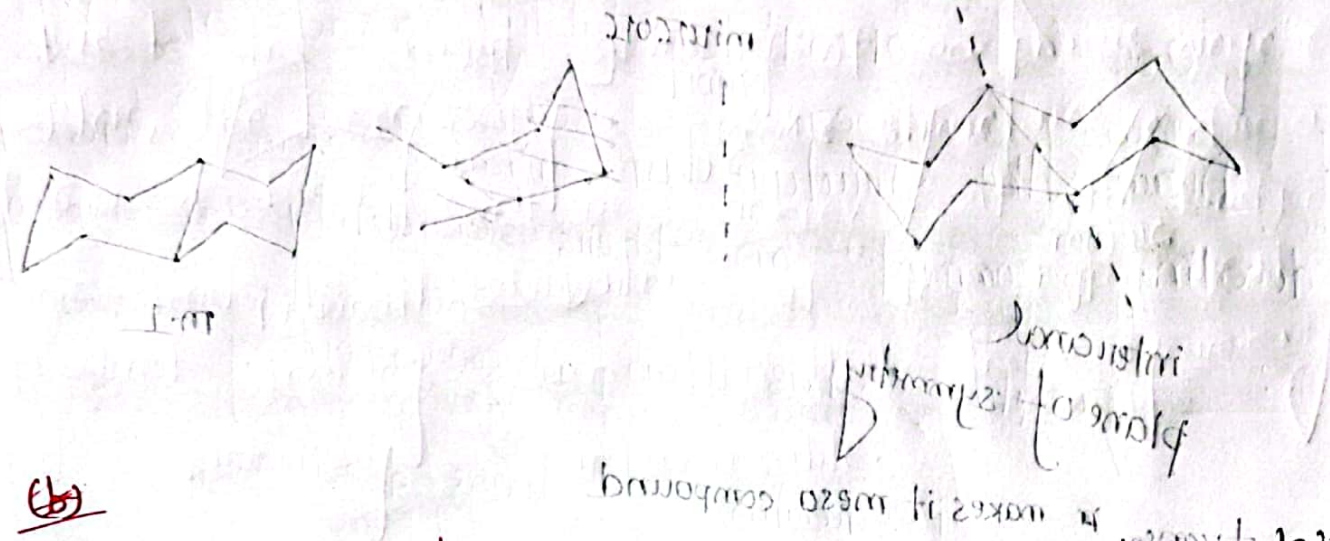
~~trans~~ ✱ makes it meso compound

(b)

(i)

(ii)

(iii)

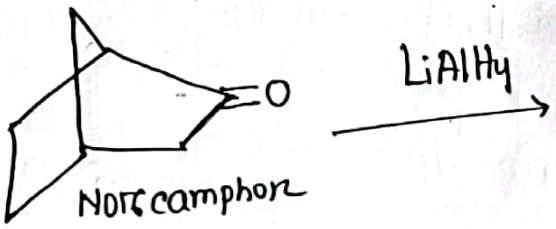


(b)

complete the following rxn and also explain the stereoselectivity of the rxn—

ANS:

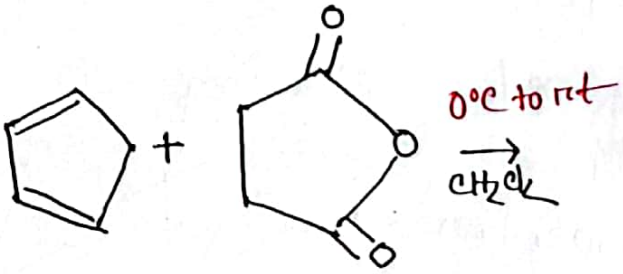
(i)

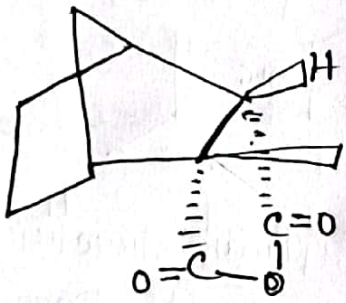


Norcamphor

endo-Norborneol

(ii)





endo product

The 2nd reaction approaches to the endo product following stereoselectivity. This rxn accompanied with diels alder reaction and endo product is ~~formed~~ often favoured because of a secondary orbital ~~si~~ interaction between the dienophiles substituent and the dien's π system during the T.S., stabilizing the endo transition state making the product more stereoselective.

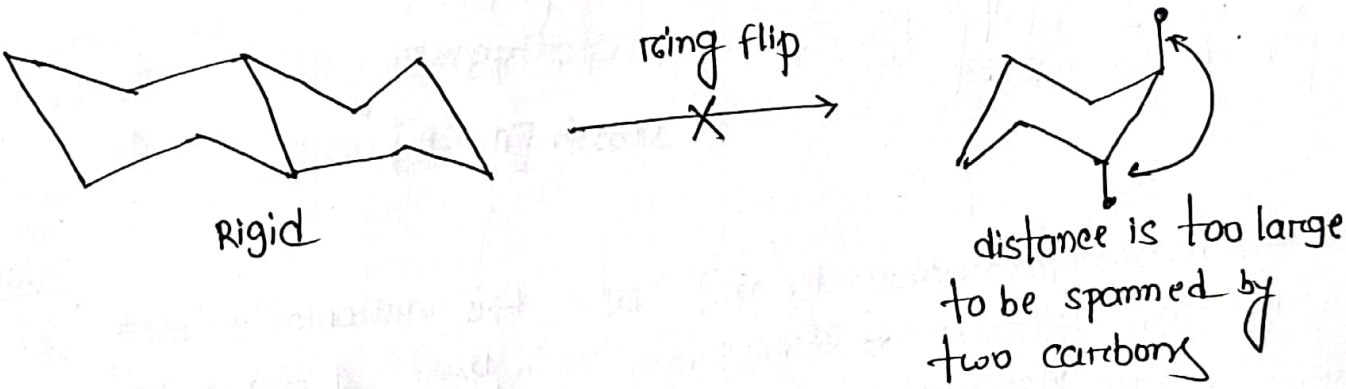
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In reaction (I) the endo side of the norcamphor is more sterically crowded than exo side, so the exo side is more accessible to HO⊖ (Nu) than the endo side.

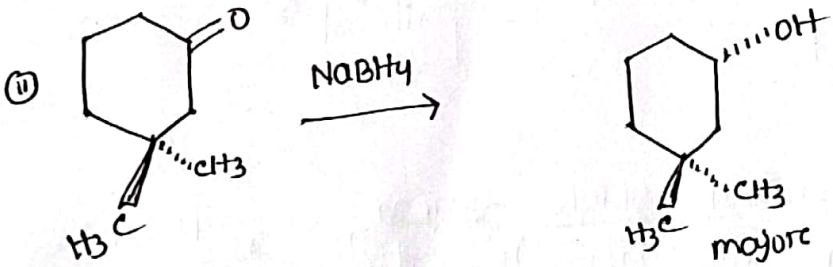
2020

Q. Why is trans decalin unable to undergo a ring flip ?

ANS: It is due to its rigid structure of trans decaline also both of its fused rings are locked in a stable conformation due to the nature of the trans-fusion at the ring junction

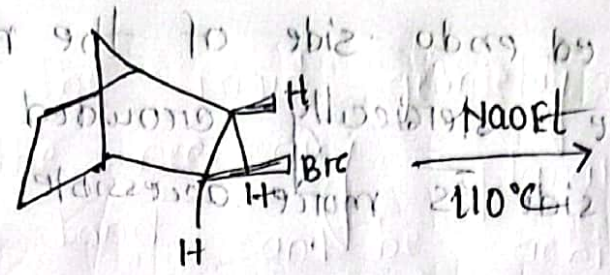


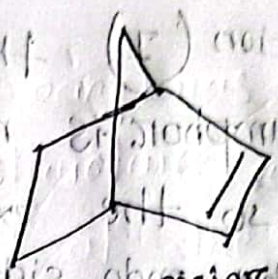
(II)



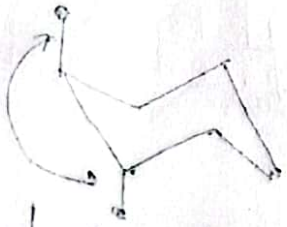
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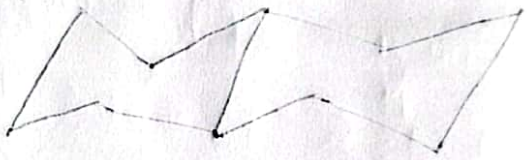
#





Explanation— Done

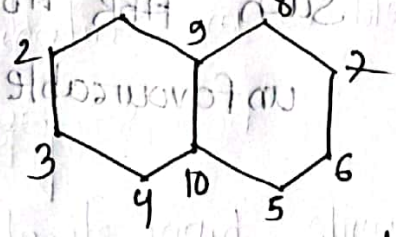




bicyclo

Draw the planar structure of decalin with the numbering of all carbon atoms and answer the following question.

Ans:



planar structure of decalin

2016

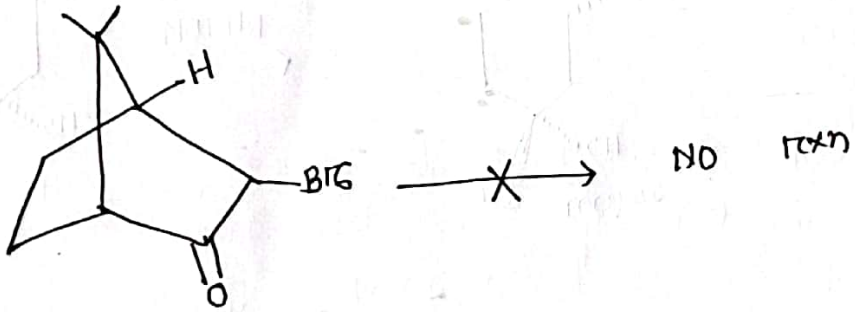
- This is the fused compound
- Another two names of decalin :

1. Decahydronapthalene
2. Bicyclo [4.4.0] decane

2016

The elimination of HBr from bromocamphor in the presence of a base isn't possible – Justify the statement on the basis of the Bredt's rule–

Ans:

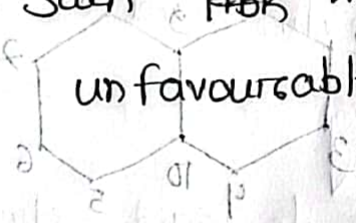


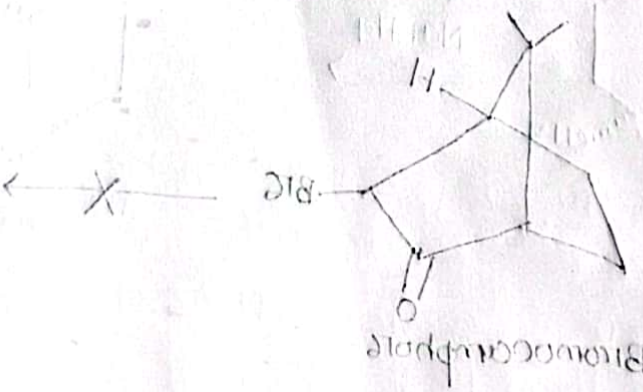
Bromocamphor

According to Bredt's rule · A bridged bicyclic compound can't have a double bond at a bridgehead position unless one of the ring contains at least 8 carbon atom.

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Due to the rigid geometry of camphor's bicyclic system, the necessary planar ~~structure~~ conformation for a π bond can't be achieved at the bridgehead. This leads to destabilization of the hypothetical product making such ~~HBr~~ HBr ~~thermodynamically~~ and kinetically unfavourable.





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